

Rebuilding Molecular Motors towards Programmable Viral DNA Packaging

A single-chain, genetically-addressable ϕ 29 gp16 packaging ring — designed and evaluated end-to-end in silico

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This report is the detailed companion to the three-minute deck. It follows the same logical arc — state the problem, redesign the motor as one addressable chain, then evaluate that design statically and dynamically — and adds the full record of what was tried, what worked, what did not, and how each claim is bounded. Everything reported here is computational. Cross-predictor agreement is treated as a filter, not a proof, and the physics-based results are driven models rather than measurements; those boundaries are stated where each result appears rather than hidden.

The strongest motor in biology is also the least engineerable

The bacteriophage ϕ 29 gp16 ATPase is among the most powerful molecular motors known. It packages a viral genome into a pre-formed capsid against forces of roughly 57 pN, pressurizing the shell to tens of atmospheres. It does this as a ring of five identical subunits that hydrolyze ATP in a coordinated cycle to translocate double-stranded DNA (Chistol 2012; Tafoya 2018; cryo-EM structure PDB 7JQQ).

That symmetry — five copies of one subunit — is exactly what makes the motor hard to interrogate and hard to engineer. Three obstacles follow directly from it, and they are the logical spine of this project.

- Reprogramming is blocked. A mutation introduced into a homo-oligomer lands at a random, unknown seat. One cannot ask what a defect at a chosen position does, nor deliberately tune or upgrade a single subunit.
- Mechanism is blocked. Single-molecule and structural readouts cannot tell identical subunits apart, so the firing order — sequential / hand-over-hand versus concerted — and the identity of any “special” regulatory subunit remain ambiguous.
- Computation is blocked. Structure predictors and monomer-scale simulations are built around single chains. A five-body homo-oligomer whose catalysis is shared across subunit interfaces is awkward to fold, score, and perturb; and five near-identical copies bleed evolutionary signal across each other, so global confidence cannot separate a correctly ordered ring from a compact but scrambled one.

The stakes follow from the obstacles. If a single subunit could be addressed, the strongest packaging machine in biology would become both readable — its mechanism resolvable one seat at a time — and tunable — its activity adjustable position by position. That is the aim: a concrete step toward programmable viral DNA packaging, with relevance to phage engineering, viral-vector production, and DNA-nanotechnology packaging.

The idea: rebuild the ring as one addressable chain

A single move removes all three obstacles at once — covalently fuse the five subunits into one gene, so the oligomer becomes one protein. A single chain makes each seat individually addressable, makes the ring computable as a monomer, and yields a defined molecular scaffold for downstream experiments. Single-chain covalent oligomers are preceded — single-chain ClpX is the closest example — but had not been built for gp16.

The work then has two halves, both carried out entirely in silico. The first is design: rebuild the five-subunit ring as one covalent single chain that folds and catalyzes like the native motor. The second is evaluation, along two axes — static (does the chain keep the native machinery: the fold, the grip on DNA, the channel, the catalytic coupling?) and dynamic (does it run the packaging cycle, and what actually drives DNA translocation?). The evaluation is what turns a redesigned protein into a tested hypothesis, and it is deliberately the larger half of the effort.

Designing the single chain

A topology-escalation ladder

Fusing five subunits into one chain is not a single trick but a choice among topologies, and the choice is dictated by two facts read directly off the native motor: whether a chain terminus fouls the DNA channel, and how far apart the subunit termini sit. Three construction strategies were compared, tried in increasing order of invention: direct end-to-end fusion, circular permutation, and de-novo diffusion linkers. Each candidate was folded by at least two independent predictors (the free Boltz-2 and OpenFold3 NIM services) with AlphaFold3 as a third, tie-breaking check, and scored on ring-context interface geometry rather than on global confidence.

Strategy	Best construct	Boltz-2	OpenFold3	AlphaFold3	Outcome
Direct N–C fusion	pentamer L30, ~1,755 aa	5/5	5/5	0/5, scrambled	rejected by AF3
Circular permutation	cp233, 1,750 aa	5/5	5/5	5/5	selected
RFdiffusion de-novo linker	salami ring, 1,835 aa	5/5 (tiled)	split	2/5, scrambled	set aside

Why direct end-to-end fusion fails

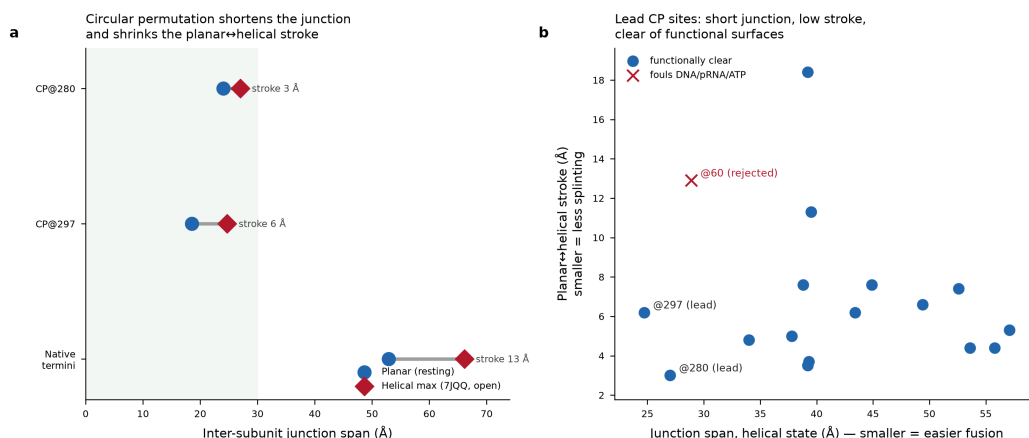
The most obvious construction — join each subunit’s C-terminus to the next subunit’s N-terminus with a flexible linker — fails for a specific, geometric reason. In the native ring the head-to-tail termini sit 54–66 Å apart, right beside the DNA channel, and the native C-terminus (residue 330) is itself a DNA-contact residue only about 6 Å from the duplex. A direct tether across that gap both fouls the channel and is too long to enforce a unique subunit order.

The failure is subtle because it hides from global confidence. A direct-fusion pentamer with an eight-repeat linker folded to Boltz-2 5/5 but only OpenFold3 2/5 and AlphaFold3 1/5, and all three reported nearly identical, unremarkable global confidence. The scrambled OpenFold3 model was in fact a geometrically perfect circle at higher interface confidence than the correct Boltz-2 ring. Shortening the linker removed the slack and closed the correct ring under both NIM

predictors, but AlphaFold3 still scrambled the chain at every linker length tested from twenty to fifty residues: it threaded the chain into a compact, planar, but mis-ordered ring in which the designed neighbors sit 34–55 Å apart and each arginine finger reaches into the wrong partner. This is a real ring in the wrong register. The lesson is that terminus position, not sequence homology, forbids direct fusion for gp16, and that a construct can look confident and still be wrong — which is why the whole project scores register, not confidence. The direct-fusion route was abandoned.

Why circular permutation

Circular permutation solves the geometry. Instead of tethering the fouled head-to-tail gap, it introduces a new chain break in a quiet surface loop, reconnects the native termini with a short internal bridge, and joins subunits at a relocated seam. That seam is short and mechanically compliant: where the native head-to-tail junction spans 54–66 Å and swings about 13 Å between the planar and helical states, the permuted junctions span roughly 24–27 Å and move only 3–6 Å per stroke, and they sit clear of the DNA, pRNA, and ATP surfaces. The linker is sized to the maximum stroke plus flexible excess, so it never splints the power stroke.



Why circular permutation. Native termini sit ~53 Å apart and swing ~13 Å per stroke, so a direct fusion strains and scrambles the register; circular permutation relocates the seam to a short, low-stroke loop clear of the functional surfaces.

The winning construct, cp233, is a 1,750-residue single chain of five permuted copies, each copy comprising native residues 234–330, a short internal linker, and native residues 4–233, with the disordered native termini trimmed. Under this topology the arginine finger and Walker-A motif that form the catalytic interface both sit within the same copy, and the chain order fixes the five ring positions, so every seat becomes an independently mutable sequence block.

The generative attempt, and why it was set aside

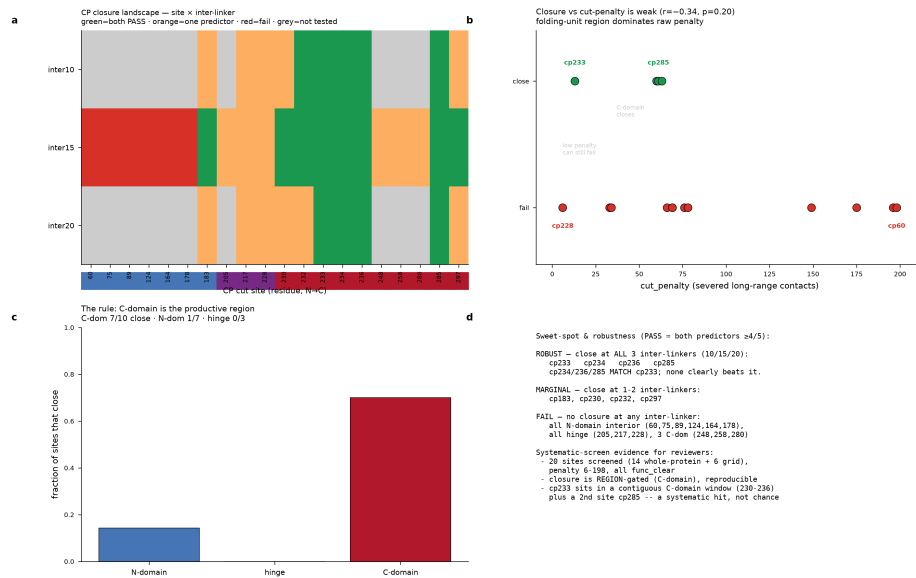
The most ambitious route — generate a de-novo connector with RFdiffusion rather than relocate the seam — was pursued and is reported here because the negative result is informative. Two modes were run. In the first, partial diffusion was used to try to rigidify cp233; across noise settings the arginine-finger contact loosened rather than tightened (from 8.1 to 8.4 Å), indicating cp233 was already near-optimal. In the second, a new connector was diffused between two adjacent subunit motifs and re-sequenced with ProteinMPNN, then tiled into a 1,835-residue ring. These rings closed into compact, sequentially ordered rings, but the inverse-folding step deleted the native arginine finger in four of five copies, and the fingers ended 40–57 Å from their designed neighbors — the topology locked while the catalytic register did not. The same sequence

was also predictor-split: clean under Boltz-2 with a tiled input but scrambled under AlphaFold3. A control confirmed that where a generative ring does close, a plain flexible linker of matched length closes it too, so the designed connector was not the load-bearing ingredient. Because cp233 was already unanimous across three predictors while the generative rings were marginal and predictor-dependent, the generative route was formally set aside — a clean negative that the framework records rather than hides.

The systematic cut-site screen

Choosing circular permutation still leaves the question of where to cut, and the answer was found by a systematic screen rather than by guessing. An early attempt ranked cut sites by the static geometry of the fusion span alone and selected a site in the middle of the C-terminal domain; the fold showed that domain fragmented and displaced by about 34 Å, with catalysis dead. That failure established the governing rule: cut only at folding-unit boundaries, and tether rather than permute through a fold.

Fig 11 — Whole-protein circular-permutation closure screen: 20 cut sites (14 whole-protein + 6 grid; penalty 6-198, all func_clear). Ring closure is gated by folding-unit region (C-domain productive), showing cp233 is a systematic hit, not chance. Free NIM, sequential MD+M1, apo, cross-checked.



The systematic cut-site screen. Twenty candidate sites, each scored on how many native long-range contacts the cut severs and on whether the ring closes across predictors; the productive region is the boundary of the C-terminal domain, and residue 233 is the lowest-penalty site. A second, independent site (cp285) also closes, showing the winner is systematic rather than incidental.

The refined screen scored roughly twenty candidate cut sites by a domain-aware penalty — the number of native long-range contacts a cut would sever, which favors loops and domain boundaries — combined with cross-predictor closure. Residue 233, at the boundary between the N-terminal ATPase domain and the C-terminal domain, carried the lowest penalty of the set, while sites buried inside a domain carried penalties several times higher. The gp16 fold divides into an ATPase domain (roughly residues 4–200, carrying the Walker-A motif, Walker-B motif, and the R146 arginine finger), a short hinge, and a C-terminal domain (roughly 229–330) that forms the DNA-gripping surface of the ring.

Choosing the linker lengths

Cut site and linker length were then screened together over a grid of three candidate sites and three internal and three inter-subunit linker lengths — twenty-seven constructs in all — each folded on a tiled input and scored for ring closure and coupling. All twenty-seven closed the

coupled ring, which shows that linker length is a second-order knob once the cut site is right; what the linker length controls is the regularity of the ring rather than whether it forms. Among cp233’s grid points, the fifteen-residue internal and ten-residue inter-subunit combination was carried forward as the lead because it is the point confirmed by all three predictors, and it became the scaffold for every downstream experiment.

Why cp233 specifically — the functional gate

Ring closure alone does not single out cp233. Two other sites, cp285 and cp297, also close the ring to 5/5 across all three predictors. What separates them is the functional gate applied below closure — whether the channel can still thread DNA — and here the three sites diverge sharply.

Construct	Cut bond	C-domain fold RMSD	Min pore diameter	Admits dsDNA	Cut severs a DNA contact
native apo (reference)	—	1.8 Å	20.6 Å	yes	—
cp233	233–234	1.9 Å	20.8 Å	yes	no
cp285	285–286	13.9 Å	14.2 Å	no	no
cp297	297–298	12.7 Å	9.3 Å	no	yes (residue 297)

The cut at 233 falls exactly on the domain boundary and spares the entire C-terminal domain, so that domain keeps its native fold and the channel stays open at 20.8 Å, matching native apo’s 20.6 Å and wide enough to thread the 20-Å duplex. The cuts at 285 and 297 land inside the C-terminal domain and shatter it, collapsing the pore to 14.2 and 9.3 Å so that neither can thread DNA; cp297 additionally severs a DNA-contact residue and drives linker atoms into the channel. cp285 is retained not as a competitor but as evidence that the win is systematic: it is a second, independent domain-boundary site that also closes, disqualified only at the functional gate. This is the reason the scoring is layered — closure is a necessary but strictly weaker test than function.

Other design axes, and the enabling trick

Several further axes were explored. The cp233 backbone was inverse-folded into a panel of de-novo re-sequenced variants sharing the fold but differing in roughly half their sequence; the permuted topology tolerated this redesign far better than the native ring did, with three of five variants still closing on the third predictor — evidence that the covalent topology, not the specific native sequence, is what makes the ring predictor-robust. A domain-recombination chimera and a multi-piece “cut-splice” topology were also specified; the cut-splice family produced an independent covalent-dimer-in-ring lead, while the chimera remained a planned rather than executed axis. Underlying all of this is one enabling method: a large single-chain ring cannot be folded from a single sequence — even the validated cp233 collapses to zero engaged interfaces under a single-sequence input — so every construct is folded with a block-diagonal, “tiled” multiple-sequence alignment that supplies each copy its own evolutionary context. That tiling is what lifts cp233 from zero to five engaged interfaces and makes the whole study possible.

The design folds like the native motor

Scoring philosophy: function-first geometry, never global confidence

Before the structural results, the scoring deserves a word, because it is a deliberate choice. Global confidence scores such as pTM and pLDDT are never used as a pass criterion. The reason is empirical: in this system a scrambled, mis-ordered ring can carry higher global and interface confidence than the correct one, and a generative ring with the single highest interface confidence in the whole study had the worst catalytic coupling. Confidence and catalytic correctness are decoupled for repeat rings. Every construct is therefore judged on ring-context interface geometry, grounded on the experimental 7JQQ structure.

The five-level metric hierarchy

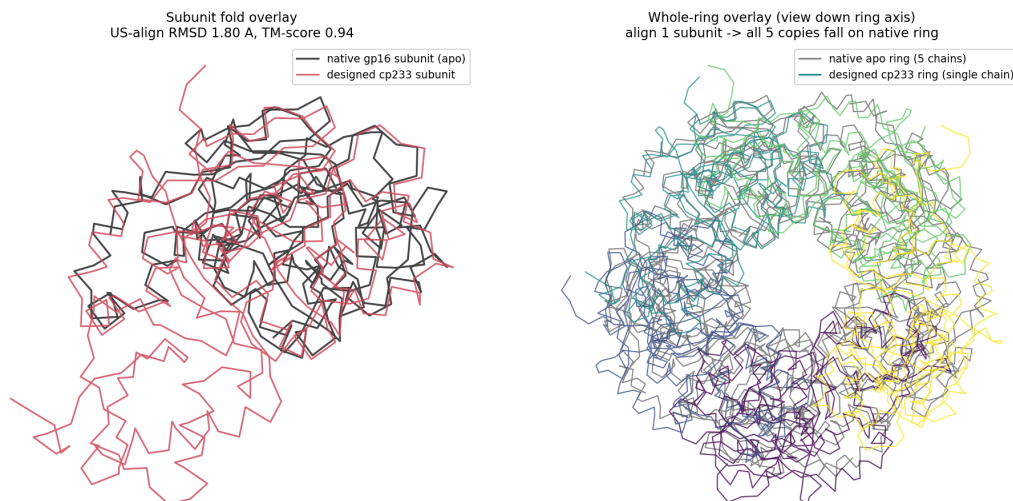
Five geometric metrics are applied in order of priority. M1 asks whether the chain forms a closed, compact, planar ring whose subunits are threaded in the designed cyclic order — the last clause is what catches a perfect circle built in the wrong register. M2, the decision metric, asks whether each subunit's trans arginine finger (R146) reaches into the neighbor's Walker-A ATP pocket, the hallmark of inter-subunit coupling; it is measured as a minimum heavy-atom distance with an 8-Å engagement threshold and a pass at four of five interfaces, and it is scored "handedness-robust" — reading the ring in both winding directions and taking the coherent maximum — because a near-symmetric ring samples left- and right-handed windings that a direction-specific score would misread. M3 asks whether the central channel can thread dsDNA and whether the DNA-contact residues line the pore; M4 and M5 check the pRNA-binding surface and the ATP pocket. The two functional souls of a packaging motor are M2 (it couples) and M3 (it threads DNA), and the functional contacts for M3–M5 are taken from the 7JQQ complex.

Structural identity to native

By these measures cp233 is indistinguishable from the native ring. A single designed subunit superimposes on the native apo subunit at 1.80 Å RMSD and a TM-score of 0.94 — the hardest, predictor-independent comparison, made against the experimental structure — and aligning one subunit places all five copies onto the five native positions. The three independent predictors agree on all five catalytic interfaces, and an independent adversarial re-scoring of the folds, recomputing the geometry from the coordinates, confirmed every one. The global confidence of these correct folds is unremarkable (pTM around 0.39), which is precisely the point: the metric that matters passes while global confidence looks mediocre.

A predictor-independent physics check

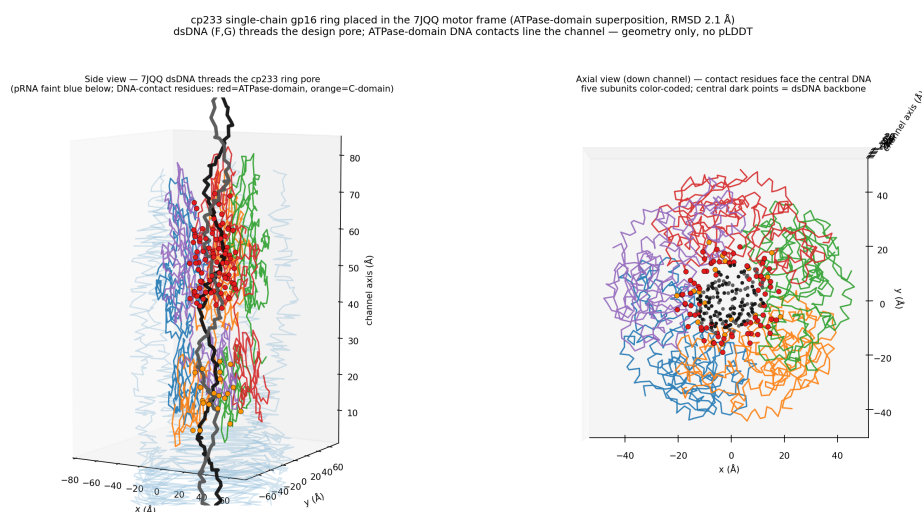
Because the three predictors share training data and priors, their agreement is a filter rather than an independent proof, so the design was also tested with physics that knows nothing of them. Under identical implicit-solvent molecular dynamics, the apo native ring stays closed, the ATP-bound helical native state (with ligands stripped) stays open with a persistent wide seam, and the design tracks the apo state — it stays closed and, over the run, becomes slightly more symmetric rather than opening. The pre-registered kill criterion was that the design would be unreasonable if it opened or collapsed while the apo ring stayed closed; it did not. Because the native motor is known to visit a helical intermediate during translocation, the design's preference for the closed planar state is a readout against a real mechanistic axis, not generic stability. This is a stability screen on short trajectories, not a converged free-energy statement, and it is described as such.



The design is structurally indistinguishable from native. Left: one designed subunit overlaid on the native subunit, 1.80 Å RMSD and TM-score 0.94. Right: all five copies of the single chain fall onto the native ring.

The design works like the native motor

Structure is the weaker test; function is the real one. Placed in the native motor frame with modeled dsDNA, the design threads the double helix through its central channel and lines the pore with its DNA-contact residues at the native positions. The rival cuts that merely close the ring cannot do this — their channels are too narrow — which is why closure is never accepted as success on its own.



The design is functionally faithful. In the native motor frame, the design's central channel threads dsDNA and grips it at the native contact residues; side and axial views.

The functional scoring is grounded on the experimental complex: twenty DNA-contact residues and twenty-nine pRNA-contact residues define the surfaces the design must preserve, and the ATP pocket is checked against the ligand-bound native geometry. cp233 matches native on all of them — the DNA-contact residues face inward at native-like radii, the ATP-pocket geometry is within a fraction of an Ångström of native across all five subunits, and the cut and trimmed termini touch no DNA or pRNA contact. The design is scored on these function-first measures throughout, never on global confidence, which makes it a defined and falsifiable scaffold for the bench rather than a confident-looking guess.

Dynamics: does it run, and what moves the DNA

The static picture says the design is a faithful copy at rest. The dynamic question is whether it runs the packaging cycle and what drives translocation, and it was studied with driven all-atom molecular dynamics on a 31,578-atom system — the single chain threaded with a 60-base-pair duplex — on cloud GPUs. One caveat frames everything in this section: these are driven models. A collective variable forces each transition roughly ten million times faster than the real millisecond-scale motor, so no rate and no absolute free energy is claimed. What is defensible is that the cycle runs, that coupling survives it, the geometry of the per-subunit motion, and the mechanistic logic the models expose.

The driven planar-to-helical cycle

Driven along a staircase collective variable, the ring ascends from the planar state into the helical lock-washer and returns to planar reversibly, across three forward and three reverse trajectories; the covalent chain is not trapped in the helical basin. Two features are robust. First, the catalytic coupling survives both half-strokes — the arginine-finger contacts do not systematically break as the ring goes up or comes down — which is the load-bearing single-chain result, since the covalent chain holds subunits coupled through the whole cycle in a way a free oligomer can shed. Second, the per-subunit motion is a graded, unequal staircase rather than five identical steps: the treads measure about -2.2, -1.2, 0.0, +1.3, and +2.2 Å, with the middle subunit a near-stationary hinge, and the run-to-run scatter is under 0.1 Å, so the pattern is set by ring geometry, not thermal noise. This unequal-step signature is what a single-molecule experiment would look for.

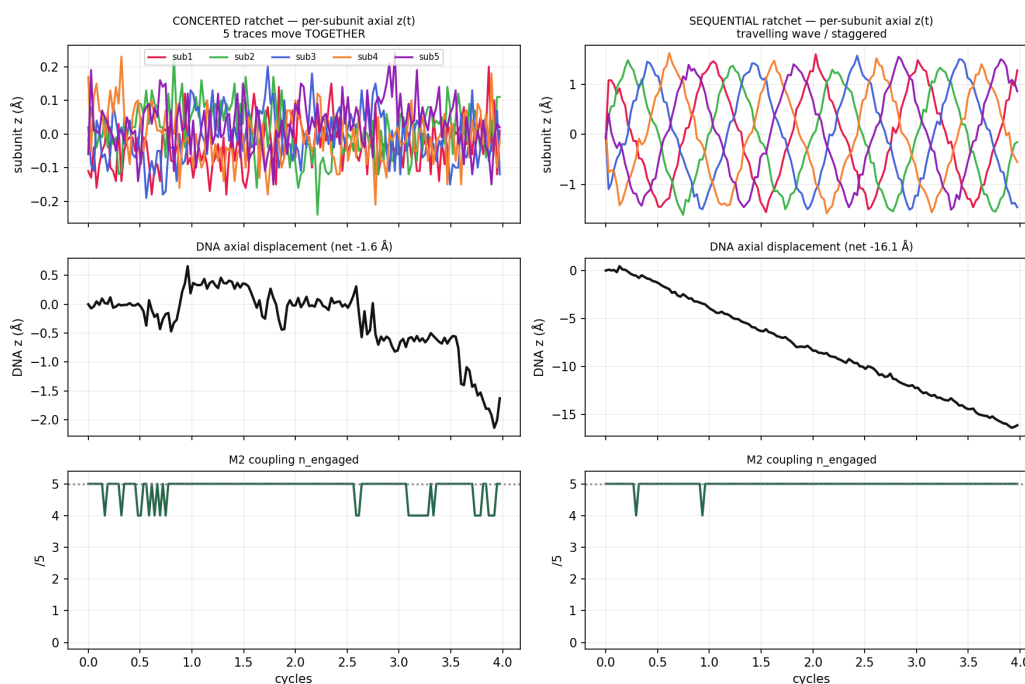
The honest translocation null

The conformational cycle, on its own, does not move the DNA. Averaged over three seeds, the net axial displacement of the duplex is statistically indistinguishable from zero in both half-strokes, with error bars larger than the means — the DNA wobbles by about an Ångström as the ring strokes around it but does not advance. One early seed happened to show a positive displacement and looked like translocation; the three-seed average erased it. This null is reported plainly, because it is not a failure but the result that motivates the next step: conformational cycling is necessary but not sufficient, and the missing element is a grip that couples ring motion to the DNA backbone.

The ratchet model, and the sequential-versus-concerted result

To ask what a grip would do, a minimal mechanochemical model added two ingredients the plain drive lacked: a per-subunit power stroke on an ATP clock, and a catch-and-carry grip that holds the DNA during a subunit's downstroke and releases on the upstroke. The model is explicitly imposed, not emergent, and its purpose is to expose a logic and a signature rather than to measure the real motor. Its central comparison is clear: sequential, hand-over-hand firing translocates the DNA by about 16 Å over four cycles, near 1.2 base pairs per cycle, while concerted firing barely moves it (about 1.6 Å). The reason is that sequential firing always keeps at least one subunit gripping, so the DNA is held continuously and cannot slip back, whereas concerted firing releases every grip at once each cycle and the DNA springs back.

This yields a falsifiable prediction that a microscope can test. Reading the phase of the five per-subunit axial traces, a travelling wave implies sequential firing and DNA substeps, while in-phase motion implies concerted firing; three-dimensional MINFLUX has the spatial and temporal resolution to make that distinction directly.



What moves the DNA. In an imposed grip-and-clock model, sequential hand-over-hand firing shows a travelling wave across the five subunits and translocates DNA about 16 Å over four cycles, while concerted firing moves in phase and barely translocates. This is a model of the mechanism and its signature, not a measured rate.

What the models together imply

A second model sharpened the conclusion in an instructive way. When the grip was placed into the symmetric helical-to-planar cycle itself, using real DNA-contact residues, translocation was small and sequential and concerted became similar — because a symmetric staircase pulls the DNA in both axial directions at once, and the two nearly cancel. Efficient translocation therefore required not just a grip but a broken symmetry, which the hand-over-hand model supplied and the symmetric cycle did not. The composite statement — conformational cycling needs a grip, and the grip needs a directional, symmetry-breaking element — is the sharpest mechanistic claim of the dynamic work, and it points to a symmetry-breaking “special” subunit as the natural phase-setter. It remains a model.

Free-energy landscape and ensemble sampling, bounded

Two further efforts are reported with their limits. Umbrella sampling along the staircase coordinate produced a free-energy profile whose shape is informative — planar is a deep basin and the helical state is strongly uphill, consistent with the design not going helical without a drive — but whose absolute numbers are not trustworthy, because the restraint was too stiff and adjacent windows did not overlap enough; the absolute barrier is a sampling artifact and is not interpreted. Separately, an equilibrium ensemble generator applicable only to monomers (and thus usable here precisely because the design is a single chain) did not reproduce the assembled ring: the samples expanded and the subunits separated, diverging from both the predictors and the unbiased dynamics. This is recorded as an honest, neutral observation — the tool does not fold large artificial single-chain rings — rather than resolved.

An interface-coupling comparison

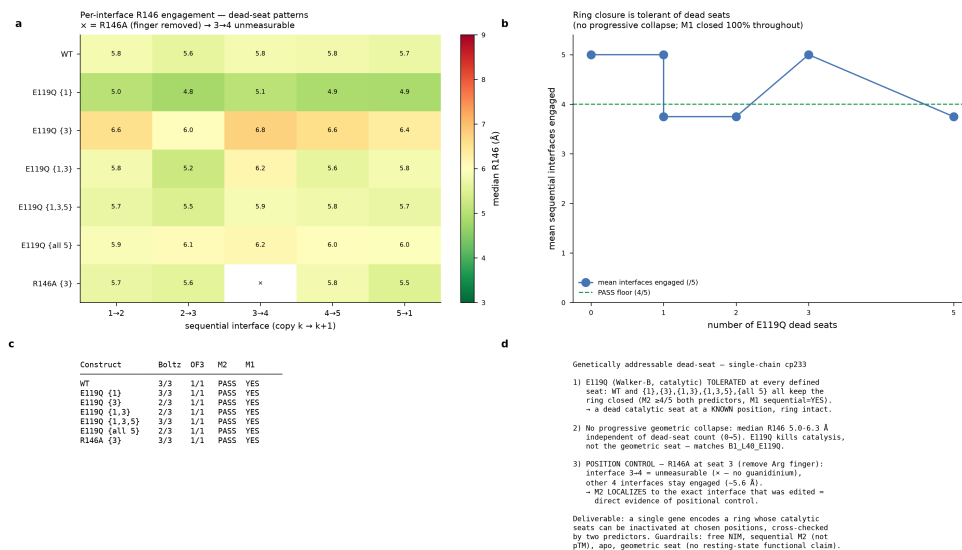
A network analysis comparing the design and native rings under identical parameters found that the covalent design transmits an active perturbation between neighboring ATP sites about 1.8 times more strongly than the native ring, and propagates it further around the ring, with a consistent force conduit through a central tyrosine hub. This gain is specific to gp16's optimized permuted topology: run on ClpX, the same comparison returns parity. It is a computed coordination proxy, not a measured force, and the one metric that favors native — passive thermal co-fluctuation — is reported alongside it. The number is a prediction for a single-molecule experiment to adjudicate, not a claim of measured enhancement.

Programmable, per-seat control

An addressable dead seat and a finger control

Because the chain is a single molecule, a mutation lands at a chosen seat. Two complementary edits demonstrate the addressability. A catalytically dead Walker-B mutation (E119Q) installed at any chosen seat, or at all five, leaves the ring fully closed and coupled — it is assembly-tolerant — and a structural analysis shows it does not distort the pocket or displace the catalytic magnesium, so it is a structurally silent, position-specific catalytic knockout whose effect is chemical and invisible to a static predictor. The arginine-finger mutation R146A is the complementary control: it disables exactly the edited interface and no other, removing one coupling contact per dead seat while the ring stays compact.

Fig 12 — Genetically addressable dead-seat on single-chain cp233_int15_inter10: Walker-B E119Q tolerated at defined seats (ring stays closed, no progressive collapse); R146A position-control localizes M2 to the edited interface. Free NIM, sequential M2+M1, apo, cross-checked.



Per-seat control. A catalytically dead seat can be installed at any chosen position while the ring stays closed, and an arginine-finger control disables exactly the edited interface — the per-seat handle that the identical native ring never allowed.

No statically special seat

Placing the arginine-finger control at each of the five seats in turn produced the same result at every position — each removal cost exactly one coupling contact, the ring stayed compact and in-order, and the damage did not propagate. By this static readout there is no structurally special seat. Read together with the dynamic result that translocation needs a symmetry-breaking

element, this implies that the “special” subunit, if it exists, is a dynamic timing feature rather than a structural one — a candidate phase-setter for the sequential wave, and a concrete target for a mixed-ring or MINFLUX experiment that the single chain now makes it possible to place at a defined position.

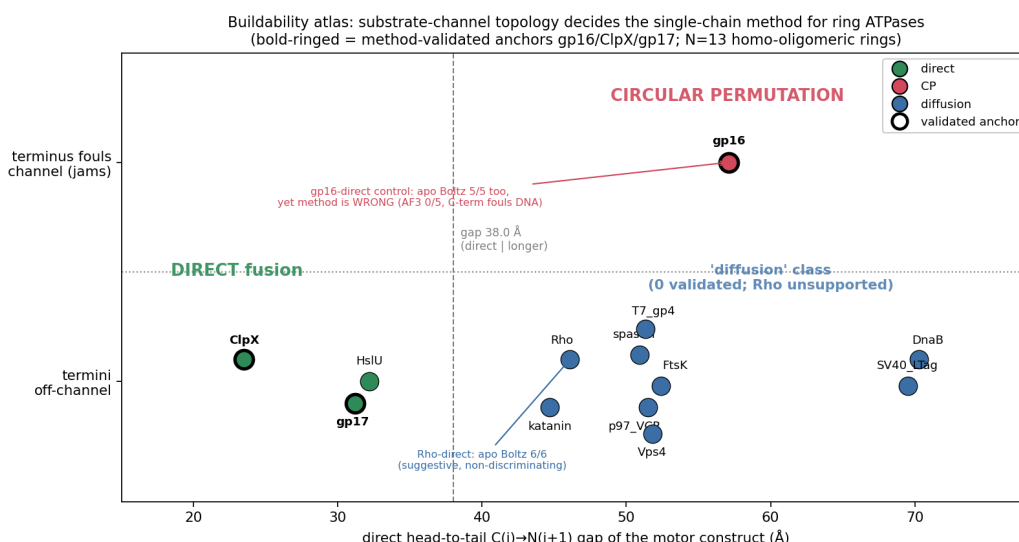
Can it be made stronger?

Whether the motor can be made stronger was tested by an in-silico directed-evolution loop that mutates, gates on structure and biochemistry, scores force- and coordination-transmission proxies, and ranks. Across two rounds the honest answer is incremental. Single mutations moved a force-transmission proxy reproducibly above the noise floor — a first-round winner and an independent second-round site rigidifying a different part of the force conduit — but coordination never cleared its noise floor, and stacking the winners into double mutants did not add: every attempted combination fell below the best single mutation. The interpretation is a rigidification ceiling: the scaffold is already near-optimal for stiffness, and a soft-mode gate rejects over-rigidified variants as dead bricks rather than stronger motors. A stronger motor would need a different axis — backbone redesign that opens new soft modes — not more stiffness. The two robust single-site variants are handed forward as concrete single-molecule predictions.

The method generalizes

A buildability rule across thirteen ring motors

The transferable claim is the method, not any single construct. Reading two descriptors off a native ring — whether a terminus fouls the substrate channel, and the size of the inter-subunit gap — sorts thirteen ring ATPases into build strategies: a terminus that fouls the channel calls for circular permutation, otherwise direct fusion sized to the gap. The rule is correct on the three systems with an independently validated answer (gp16 to permutation, ClpX and gp17 to fusion), and among the whole panel gp16 is the unique channel-fouler, which is exactly why it needs permutation while the others do not.



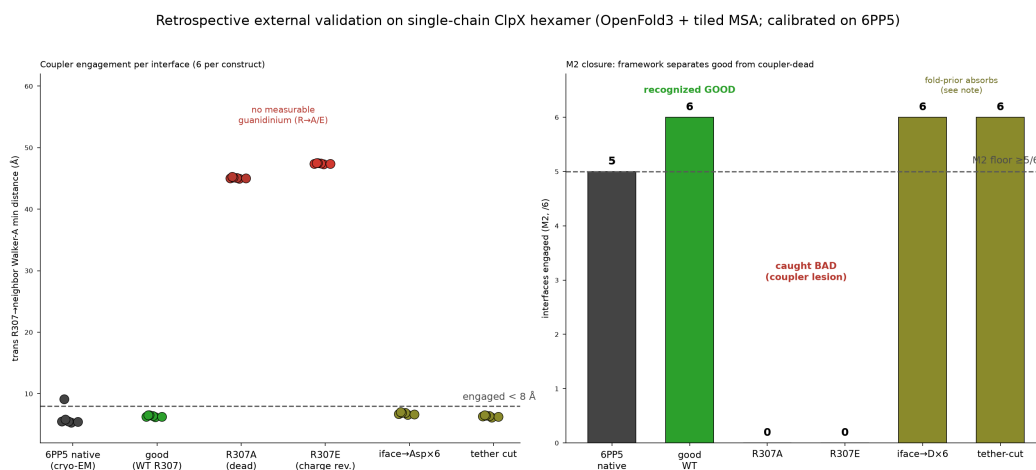
A buildability rule across thirteen ring motors. Two descriptors read off the native ring separate the systems that need circular permutation from those that fuse directly; gp16 is the unique channel-fouler.

An important cautionary result is reported alongside: an apo, protein-only fold cannot by itself adjudicate the build choice, because a known-wrong direct-fusion gp16 still closes a clean ring

when the clashing DNA channel is absent from the input. A single fold is therefore blind to substrate-channel fouling, and the honest rule collapses to two branches — a terminus that fouls the channel needs permutation, everything else fuses. It passes the healthy systems and flags the one broken case.

A blind retrospective test on ClpX

Finally, the fold-in-context framework was run blind on a second, unrelated ring ATPase — the AAA+ unfoldase ClpX, built as a single-chain pseudohexamer. Calibrated on the experimental ClpX structure, where it reproduces the native pattern, the coupling metric passes the working single-chain construct at six of six interfaces and flags the known dead-coupler arginine-finger mutants at zero of six.



A blind retrospective test on ClpX. The coupling metric passes the working single-chain construct 6/6 and flags the known dead-coupler mutants 0/6, showing the framework transfers across the AAA+ family.

Two other broken constructs — a severed tether and scattered surface mutations — still closed, which is informative rather than a failure: the metric is specifically a coupler-engagement read-out, catching the arginine-finger lesion that defines the trans catalytic contact, not a general assembly oracle. That the discriminating axis is the same catalytic coupler as in gp16 is the point. The transfer is demonstrated on prediction and native-structure calibration, not on wet-lab data.

What this changes

Turning a homo-oligomeric ring into a single, genetically addressable chain converts a machine that could not be interrogated or engineered into one that can, and it does so with three concrete consequences. It provides a per-seat perturbation handle — the substrate for asking which subunit, if any, is special, and for tuning one position without touching the others. It yields a falsifiable mechanism prediction — efficient translocation requires sequential firing with a continuous, symmetry-broken grip, detectable as a travelling wave under three-dimensional MINFLUX. And it delivers a transferable design method — one rule builds single chains across a family of ring motors. Together these begin to make the strongest DNA-packaging machines in biology both readable and tunable.

Limitations

The boundaries of the work are stated plainly and are the reason nothing here is described as validated. Everything is computational; no experiments have been run. The structure predictors

are not independent, sharing training data and priors, which is why physics-based dynamics is used as the orthogonal check and cross-predictor agreement is treated as a filter. The dynamics are driven models rather than spontaneous trajectories: the real stroke is far too slow to simulate, the ratchet's grip and clock are imposed, and the plain-cycle DNA translocation is an honest null. The free-energy landscape is under-sampled and used for shape only. The coupling gain is specific to gp16 and vanishes on ClpX, and a computed coordination proxy is not a measured force. An explicit-solvent force measurement was attempted and deferred when the large solvated system proved numerically unstable, and even a successful pull would report a drag force rather than the physiological stall. The natural next steps are exactly the experiments the design was built to enable: single-molecule measurement with optical tweezers and three-dimensional MINFLUX, ATPase biochemistry, and cryo-electron microscopy of the single-chain ring.

Reproducibility and how Claude was used

The scorers, folding pipelines, molecular-dynamics drivers, and raw outputs, together with the full commit history, are in this repository under `gp16_design/`. The compute was orchestrated with Claude Code — GPU jobs on Colab, cloud A100s, and NVIDIA NIM services through hard-timeout harnesses, alongside a multi-agent workflow that scored, cross-checked, and synthesized every fold and surfaced the honest nulls — while the systematic folding and validation campaigns ran in Claude Science. The total compute was under \$50 of GPU time across 94 commits, with the design and scoring logic packaged as a reusable toolkit.